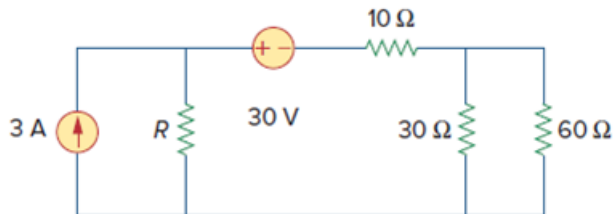


## Problem

Consider the  $30\text{-}\Omega$  resistor in Fig. 4.134. First compute the Thevenin equivalent circuit as seen by the  $30\text{-}\Omega$  resistor. Compute the value of  $R$  that results in Thevenin equivalent resistance equal to the  $30\text{-}\Omega$  resistance and then calculate power delivered to the  $30\text{-}\Omega$  resistor. Now let  $R = 0\text{ }\Omega$ ,  $110\text{ }\Omega$ , and  $\infty$ , calculate the power delivered to the  $30\text{-}\Omega$  resistor in each case. What can you say about the value of  $R$  that will result in the maximum power that can be delivered to the  $30\text{-}\Omega$  resistor?

Figure 4.134:



## Step-by-step solution

## Step 1 of 10

Refer to Figure 4.134 in the textbook.

- Calculate Thevenin resistance,  $R_{TH}$ .

Replace the  $3\text{ A}$  current source by open circuit and the  $30\text{ V}$  voltage source by the short circuit. Also, replace the  $30\text{ }\Omega$  resistance by the open circuit to calculate the Thevenin resistance, as shown in Figure 1.

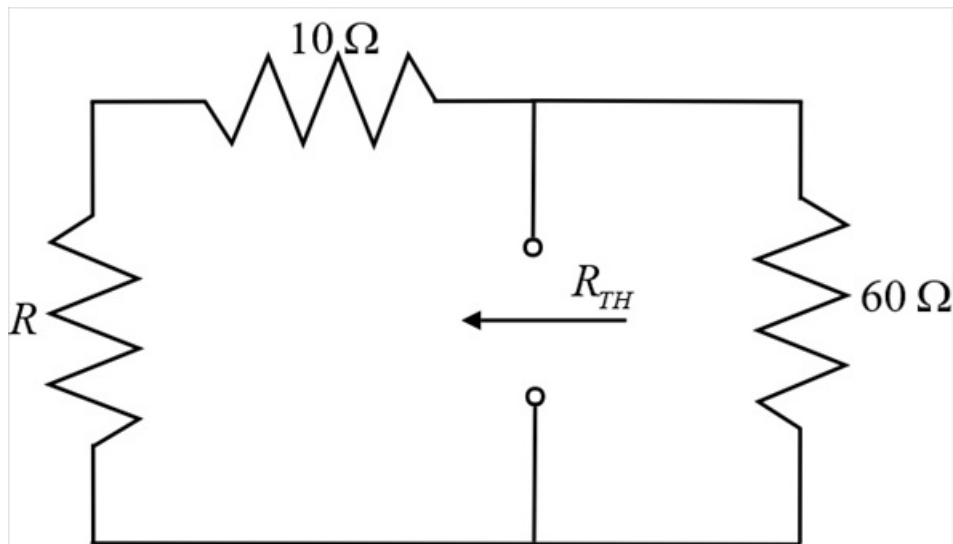


Figure 1

## Step 2 of 10

The resistance  $R$  and  $10\ \Omega$  are in series. Then resistance  $R_1$  is as follows:

$$R_1 = R + 10$$

Resistance  $R_1$  and  $60\ \Omega$  are in parallel. Then, Thevenin resistance  $R_{TH}$  is as follows:

$$\begin{aligned} R_{TH} &= (R_1) \parallel (60) \\ &= (R + 10) \parallel (60) \\ &= \frac{(R + 10)(60)}{(R + 10 + 60)} \\ &= \frac{(R + 10)(60)}{(R + 70)} \dots\dots (1) \end{aligned}$$

### Step 3 of 10

- Calculate the Thevenin voltage,  $V_{TH}$ .

Refer to Figure 4.134 in the textbook.

Convert the  $3\text{ A}$  current source into voltage source by multiplying the current and the resistance  $R$  in parallel.

$$\begin{aligned} V_{con} &= (3)(R) \\ &= 3R \end{aligned}$$

Replace the parallel combination of  $3\text{ A}$  current source and resistor  $R$  with the series combination of the  $V_{con} = 3R$  voltage source and resistor  $R$ . Replace the  $30\ \Omega$  resistor by the open circuit, as shown in Figure 2.

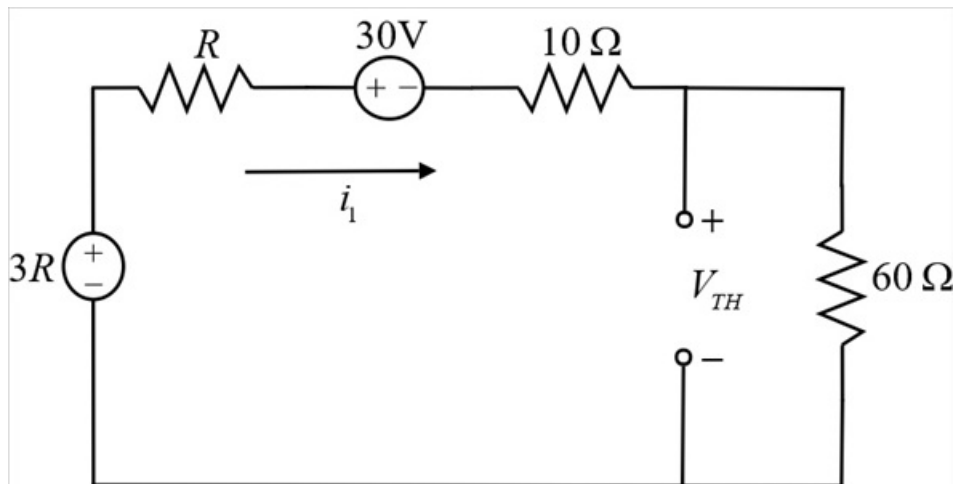


Figure 2

### Step 4 of 10

Apply Kirchhoff's voltage law in the circuit.

$$-3R + i_1 R + 30 + 10i_1 + 60i_1 = 0$$

$$3R = i_1 R + 30 + 10i_1 + 60i_1$$

$$3R - 30 = i_1 (R + 10 + 60)$$

$$i_1 = \frac{3R - 30}{R + 70}$$

Calculate the Thevenin voltage,  $V_{TH}$ .

$$V_{TH} = i_1 (60)$$

Substitute  $\frac{3R - 30}{R + 70}$  for  $i_1$ .

$$V_{TH} = \left( \frac{3R - 30}{70 + R} \right) 60 \dots\dots (2)$$

**Step 5** of 10

The Thevenin equivalent circuit is,

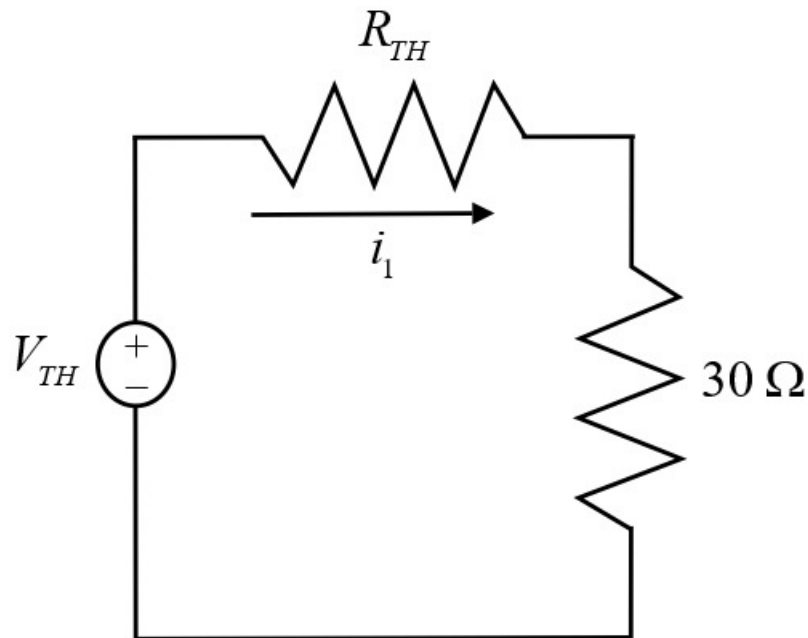


Figure 3

**Step 6** of 10

The maximum power is transferred when the load resistance ( $R_L$ ) is equal to the Thevenin resistance ( $R_{TH}$ ).

$$R_{TH} = R_L$$

Since the load resistance is  $R_L = 30 \Omega$ , therefore,

$$\begin{aligned} R_{TH} &= R_L \\ &= 30 \Omega \end{aligned}$$

Substitute  $30 \Omega$  for  $R_{TH}$  in equation (1).

$$\begin{aligned} \frac{(R+10)(60)}{(R+70)} &= 30 \\ (R+10)(60) &= (30)(R+70) \\ 60R+600 &= 30R+2100 \\ (60-30)R &= 2100-600 \\ R &= \frac{1500}{30} \\ &= 50 \Omega \end{aligned}$$

Hence, the value of resistance  $R$  is  $50 \Omega$ .

#### Step 7 of 10

Substitute  $50 \Omega$  for  $R$  in equation (2).

$$\begin{aligned} V_{TH} &= \left( \frac{3(50)-30}{70+50} \right) 60 \\ &= 60 \text{ V} \end{aligned}$$

Calculate the current  $i_1$  in Figure 3. Apply Kirchhoff's voltage law.

$$-V_{TH} + R_{TH}i_1 + 30i_1 = 0$$

Substitute  $-\frac{180}{7} \text{ V}$  for  $V_{TH}$  and  $\frac{60}{7} \Omega$  for  $R_{TH}$ .

$$\begin{aligned} -60 + 30i_1 + 30i_1 &= 0 \\ 60i_1 &= 60 \\ i_1 &= \frac{60}{60} \\ &= 1 \text{ A} \end{aligned}$$

The power across the  $30 \Omega$  resistor is as follows:

$$P = i^2(30)$$

Substitute  $1 \text{ A}$  for  $i_1$ .

$$\begin{aligned} P &= (1)^2(30) \\ &= 30 \text{ W} \end{aligned}$$

Hence, the power across the  $30 \Omega$  resistor is  $30 \text{ W}$ .

#### Step 8 of 10

• For  $R = 0 \Omega$

Substitute  $0 \Omega$  for  $R$  in equation (2).

$$\begin{aligned} V_{TH} &= \left( \frac{3(0) - 30}{70 + 0} \right) 60 \\ &= -\frac{180}{7} \text{ V} \end{aligned}$$

Calculate the current  $i_1$  in Figure 3. Apply Kirchhoff's voltage law.

$$-V_{TH} + R_{TH}i_1 + 30i_1 = 0$$

Substitute  $-\frac{180}{7} \text{ V}$  for  $V_{TH}$  and  $30 \Omega$  for  $R_{TH}$ .

$$\frac{180}{7} + 30i_1 + 30i_1 = 0$$

$$60i_1 = -\frac{180}{7}$$

$$\begin{aligned} i_1 &= -\frac{180}{420} \\ &= -0.428 \text{ A} \end{aligned}$$

The power across the  $30 \Omega$  resistor is,

$$P = i^2 (30)$$

Substitute  $-0.428 \text{ A}$  for  $i_1$ .

$$\begin{aligned} P &= (-0.428)^2 (30) \\ &= 5.5 \text{ W} \end{aligned}$$

Hence, the power across the  $30 \Omega$  resistor for  $R = 0 \Omega$  is 5.5 W.

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#### Step 9 of 10

For  $R = 110 \Omega$

Substitute  $110 \Omega$  for  $R$  in equation (2).

$$\begin{aligned} V_{TH} &= \left( \frac{3(110) - 30}{70 + 110} \right) 60 \\ &= 100 \text{ V} \end{aligned}$$

Calculate the current  $i_1$  in Figure 3. Apply Kirchhoff's voltage law.

$$-V_{TH} + R_{TH}i_1 + 30i_1 = 0$$

Substitute  $100 \text{ V}$  for  $V_{TH}$  and  $30 \Omega$  for  $R_{TH}$ .

$$-100 + 30i_1 + 30i_1 = 0$$

$$60i_1 = 100$$

$$\begin{aligned} i_1 &= \frac{100}{60} \\ &= 1.67 \text{ A} \end{aligned}$$

The power across the  $30 \Omega$  resistor is,

$$P = i^2 (30)$$

Substitute  $1.67 \text{ A}$  for  $i_1$ .

$$\begin{aligned} P &= (1.67)^2 (30) \\ &= 83.67 \text{ W} \end{aligned}$$

Hence, the power across the  $30 \Omega$  resistor for  $R = 110 \Omega$  is 83.67 W.

- For  $R = \infty$

Rewrite equation (2) as follows:

$$\begin{aligned} V_{TH} &= \left( \frac{3R - 30}{70 + R} \right) (60) \\ &= \frac{R}{R} \left( \frac{3 - \frac{30}{R}}{\frac{70}{R} + 1} \right) (60) \\ &= \left( \frac{3 - \frac{30}{R}}{\frac{70}{R} + 1} \right) (60) \end{aligned}$$

Substitute  $\infty$  for  $R$ .

$$\begin{aligned} V_{TH} &= \left( \frac{3 - \frac{30}{\infty}}{\frac{70}{\infty} + 1} \right) (60) \\ &= \left( \frac{3 - 0}{0 + 1} \right) (60) \\ &= 180 \text{ V} \end{aligned}$$

Calculate the current  $i_1$  in Figure 3. Apply Kirchhoff's voltage law.

$$-V_{TH} + R_{TH}i_1 + 30i_1 = 0$$

Substitute 180 V for  $V_{TH}$  and  $30 \Omega$  for  $R_{TH}$ .

$$-180 + 30i_1 + 30i_1 = 0$$

$$60i_1 = 180$$

$$\begin{aligned} i_1 &= \frac{180}{60} \\ &= 3 \text{ A} \end{aligned}$$

The power across the  $30 \Omega$  resistor is,

$$P = i^2 (30)$$

Substitute 3 A for  $i_1$ .

$$\begin{aligned} P &= (3)^2 (30) \\ &= 270 \text{ W} \end{aligned}$$

Hence, the power across the  $30 \Omega$  resistor for  $R = \infty$  is 270 W.

Hence, as the value of resistance  $R$  is increases, the power across the  $30\text{-}\Omega$  resistor increases as well.